Online Resource 1

Do AI weather models forecast extremes better than GFS? A station-based verification over China with a common initialization source, 2022-2025

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Supplementary key-result tables and station inventory

This PDF presents the existing key-result exports for the core, core21 and paired-initialization comparisons, followed by the 334-station analysis inventory. The numerical entries reproduce the verified CSV exports. No scores or statistical tests were recomputed to prepare this document.

Public code and derived data

Repository: yizou728282/nwp-ai-extremes-china-obs-public

Submission version: submission-2026-10-04

Archive: Code_and_derived_data.zip

The public release includes processing and verification code, regression tests, station inventories, full machine-readable derived result tables, method metadata, available run/provenance records and figure sources. Manuscript files are not deposited in the public repository. The repository README explains the environment and reconstruction steps.

Contents

Tables S1-S5: core comparison (pages 3-7).
Tables S6-S9: core21 sensitivity (pages 8-11).
Tables S10-S14: paired GFS/IFS initialization (pages 12-16).
Table S15: analysis station inventory (pages 17-25).

Scope of deposited material

Large forecast-point and observation caches, auxiliary map datasets, archive chunk indices, analysis cubes and case-level score arrays are excluded. The supplied scripts document reconstruction from the public sources. The Figure 4 snapshot contains display positions and categories; it is not a substitute for the omitted case-level numerical data.

Sources and interpretation

Original public data sources

AI forecasts: NOAA/CIRA AIWP archive

GFS forecasts: NOAA Open Data on AWS

Station observations: NOAA NCEI Integrated Surface Database

Daily observations: NOAA NCEI GHCN-Daily

Climate maps: Beck et al. (2023), Figshare collection

Key-result notation

Each entry is skill relative to GFS in percent, followed by an optional asterisk and [better/worse] station percentages. Positive skill favours the AI model. An asterisk means that the 95% calendar-block bootstrap interval excludes zero. Bracketed percentages identify stations significantly better or worse than GFS using the Diebold-Mariano test with Benjamini-Hochberg FDR 5%, separately for each model, reference, variable, lead and score.

The bootstrap skill uses pooled station-case loss sums. It is distinct from equal-station summary means. The FDR adjustment is not study-wide. In-sample intervals condition on the fitted IDR maps. Values are shown with the rounding already present in the key-result CSVs; full estimates and confidence limits are in the release.

Score keys

pcrps: in-sample potential CRPS.
tw1, tw2: in-sample threshold-weighted PCRPS for tails 1 and 2 in manuscript Table 1.
oos_idr: independent-period IDR CRPS.
oos_tw1, oos_tw2: independent-period IDR threshold-weighted CRPS.
oos_emos: independent-period Gaussian EMOS CRPS.
oos_emos_tw1: independent-period Gaussian EMOS threshold-weighted CRPS for tail 1.

Comparison sets

core: January 2022-August 2025, GFS-initialized AI models; independent-period calibration trains on 2022-2023 and tests on January 2024-August 2025.
core21: includes 2021, omits GraphCast; independent-period training is 2021-2023.
ic: GFS- and IFS-initialized AI runs on common initialization dates, with GFS as the forecast reference; in-sample measures only.

Version identity

Scientific pipeline: 2026-10-03-verification-v2
Method SHA-256:

6ad4b64b41a0eeb72307249c48995698c9c51b26fa89998a8f61cf271a847d9a

Table S1. 2-m temperature

Comparison set: **core**. Skill relative to GFS (%) at the selected lead times.

GraphCast Op. = GraphCast Operational; FCN v2-small = FourCastNet v2-small. All AI runs in this set use GFS initialization.

Score	Model	24 h	72 h	120 h	168 h	240 h
pcrps	FCN v2-small	+5.9* [56/20]	+6.7* [58/11]	+7.1* [57/9]	+7.0* [55/3]	+5.1* [44/1]
pcrps	GraphCast Op.	+8.0* [70/11]	+11.2* [78/4]	+11.0* [81/2]	+10.1* [83/1]	+8.5* [78/1]
pcrps	Pangu-Weather	+8.4* [69/11]	+12.4* [78/6]	+11.5* [79/3]	+8.8* [69/2]	+3.7* [25/1]
tw1	FCN v2-small	+6.9* [10/4]	+7.2* [9/1]	+8.1* [7/1]	+8.9* [2/0]	+6.0* [0/0]
tw1	GraphCast Op.	+9.4* [22/3]	+9.1* [10/0]	+9.1* [6/0]	+8.4* [3/0]	+8.0* [0/0]
tw1	Pangu-Weather	+8.9* [20/2]	+11.8* [24/1]	+11.2* [20/0]	+10.3* [12/0]	+6.4* [0/0]
tw2	FCN v2-small	+3.7* [4/4]	-1.1 [1/1]	-3.1 [0/0]	-4.5 [0/0]	-6.0 [0/0]
tw2	GraphCast Op.	+5.1* [14/2]	+6.4* [0/0]	+5.0* [0/0]	+2.4 [0/0]	+1.8 [0/0]
tw2	Pangu-Weather	+7.2* [4/1]	+10.3* [1/0]	+7.9* [1/0]	+4.8 [0/0]	+0.7 [0/0]
oos_idr	FCN v2-small	+5.2* [43/13]	+6.3* [41/9]	+6.3* [32/4]	+6.3* [29/1]	+3.4 [15/1]
oos_idr	GraphCast Op.	+7.4* [58/7]	+10.6* [64/2]	+10.2* [62/1]	+9.3* [57/1]	+7.6* [45/1]
oos_idr	Pangu-Weather	+8.1* [58/6]	+12.3* [72/4]	+11.4* [64/2]	+8.7* [54/1]	+3.4* [5/0]
oos_tw1	FCN v2-small	+5.5* [0/0]	+5.9* [0/0]	+6.5* [0/0]	+7.8* [0/0]	+5.5* [0/0]
oos_tw1	GraphCast Op.	+7.3* [0/0]	+7.7* [0/0]	+8.1* [0/0]	+7.8* [0/0]	+5.0* [0/0]
oos_tw1	Pangu-Weather	+7.4* [0/0]	+9.9* [0/0]	+9.1* [0/0]	+8.9* [0/0]	+3.8 [0/0]
oos_tw2	FCN v2-small	+5.2* [0/0]	+2.2 [0/0]	-0.3 [0/0]	-1.7 [0/0]	+0.6 [0/0]
oos_tw2	GraphCast Op.	+7.4* [0/0]	+8.6* [0/0]	+8.8* [0/0]	+5.1 [0/0]	+2.3 [0/0]
oos_tw2	Pangu-Weather	+8.2* [0/0]	+11.5* [0/0]	+10.6* [0/0]	+8.1* [0/0]	+2.9* [0/0]
oos_emos	FCN v2-small	+3.3* [40/16]	+4.1* [33/9]	+3.8* [24/4]	+4.0* [21/0]	+1.2 [3/0]
oos_emos	GraphCast Op.	+7.1* [62/6]	+9.2* [65/3]	+8.4* [63/1]	+6.9* [50/0]	+4.7* [32/0]
oos_emos	Pangu-Weather	+7.3* [64/5]	+11.6* [72/3]	+10.9* [71/2]	+8.1* [63/0]	+4.2* [28/0]
oos_emos_tw1	FCN v2-small	+7.8* [0/0]	+6.9* [0/0]	+8.8* [0/0]	+11.4* [0/0]	+8.0* [0/0]
oos_emos_tw1	GraphCast Op.	+8.3* [0/0]	+9.4* [0/0]	+10.3* [0/0]	+9.5* [0/0]	+6.2* [0/0]
oos_emos_tw1	Pangu-Weather	+7.0* [0/0]	+10.4* [7/0]	+12.8* [1/0]	+13.0* [1/0]	+8.5* [0/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_core.csv; variable key: t2.

Table S2. Daily maximum temperature

Comparison set: **core**. Skill relative to GFS (%) at the selected lead times.

GraphCast Op. = GraphCast Operational; FCN v2-small = FourCastNet v2-small. All AI runs in this set use GFS initialization.

Score	Model	Day 1	Day 3	Day 5	Day 7	Day 9
pcrps	FCN v2-small	+2.7* [34/22]	+5.1* [32/11]	+5.6* [36/4]	+4.9* [39/3]	+4.1* [33/6]
pcrps	GraphCast Op.	+8.0* [62/18]	+9.5* [66/11]	+9.2* [68/5]	+8.4* [63/2]	+7.1* [61/1]
pcrps	Pangu-Weather	+6.1* [43/20]	+8.3* [48/9]	+7.4* [48/5]	+3.4* [40/20]	+0.6 [26/16]
tw1	FCN v2-small	+1.7 [12/4]	+3.2* [6/1]	+5.4* [6/0]	+5.4* [4/0]	+4.8* [1/0]
tw1	GraphCast Op.	+2.8* [19/8]	+3.0* [5/2]	+5.5* [3/0]	+6.5* [3/0]	+7.5* [8/0]
tw1	Pangu-Weather	+4.3* [19/6]	+5.9* [13/1]	+6.3* [9/0]	+6.1* [13/0]	+4.9* [4/0]
tw2	FCN v2-small	-1.7 [3/2]	-0.1 [1/0]	+2.7 [0/0]	+3.2 [1/0]	+3.1 [0/0]
tw2	GraphCast Op.	+0.3 [5/4]	-0.1 [0/0]	+2.5 [0/0]	+4.1* [0/0]	+6.0* [0/0]
tw2	Pangu-Weather	+1.5 [8/4]	+2.7 [0/0]	+3.3 [2/0]	+3.9* [3/0]	+3.5* [0/0]
oos_idr	FCN v2-small	+3.6* [19/9]	+6.2* [27/4]	+4.5* [22/1]	+3.7* [14/2]	+1.6 [18/7]
oos_idr	GraphCast Op.	+6.4* [44/8]	+8.8* [40/3]	+7.6* [37/3]	+7.4* [38/2]	+6.6* [34/2]
oos_idr	Pangu-Weather	+5.7* [33/11]	+8.7* [39/5]	+6.6* [34/2]	+3.5* [26/7]	+0.6 [17/7]
oos_tw1	FCN v2-small	+3.7 [2/0]	+4.2* [1/0]	+5.7* [1/0]	+6.5* [2/0]	+6.5* [1/0]
oos_tw1	GraphCast Op.	+2.5 [1/0]	+3.6* [1/0]	+5.1* [1/0]	+6.3* [1/0]	+6.8* [0/0]
oos_tw1	Pangu-Weather	+4.0* [2/1]	+5.5* [1/0]	+6.7* [2/0]	+7.4* [4/0]	+4.8* [0/0]
oos_tw2	FCN v2-small	+0.8 [0/0]	+1.0 [0/0]	+3.4 [0/0]	+4.7* [0/0]	+5.3* [0/0]
oos_tw2	GraphCast Op.	+0.4 [0/0]	+1.0 [0/0]	+1.8 [1/0]	+3.0 [0/0]	+4.3 [0/0]
oos_tw2	Pangu-Weather	+1.5 [0/0]	+2.7 [0/0]	+4.3* [0/0]	+5.3* [0/0]	+2.8 [0/0]
oos_emos	FCN v2-small	+2.9* [8/4]	+4.9* [14/1]	+4.0* [8/0]	+3.5* [6/0]	+1.7 [2/0]
oos_emos	GraphCast Op.	+6.3* [53/11]	+8.4* [52/5]	+7.9* [46/2]	+7.3* [43/0]	+6.2* [36/0]
oos_emos	Pangu-Weather	+7.8* [49/3]	+10.2* [55/1]	+9.3* [53/1]	+6.5* [41/1]	+4.6* [24/1]
oos_emos_tw1	FCN v2-small	+9.8* [8/1]	+11.0* [2/0]	+12.8* [16/0]	+12.9* [12/0]	+11.1* [8/0]
oos_emos_tw1	GraphCast Op.	+7.0* [6/0]	+9.1* [6/0]	+11.7* [20/0]	+11.6* [0/0]	+9.9* [0/0]
oos_emos_tw1	Pangu-Weather	+11.8* [18/1]	+13.0* [7/0]	+13.8* [20/0]	+11.7* [13/0]	+9.4* [0/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_core.csv; variable key: tmax.

Table S3. Daily minimum temperature

Comparison set: **core**. Skill relative to GFS (%) at the selected lead times.

GraphCast Op. = GraphCast Operational; FCN v2-small = FourCastNet v2-small. All AI runs in this set use GFS initialization.

Score	Model	Day 1	Day 3	Day 5	Day 7	Day 9
pcrps	FCN v2-small	+11.1* [75/9]	+10.1* [71/7]	+8.9* [62/5]	+6.6* [50/3]	+4.5* [46/2]
pcrps	GraphCast Op.	+8.9* [75/8]	+12.1* [84/1]	+11.7* [83/0]	+10.0* [78/0]	+8.4* [71/0]
pcrps	Pangu-Weather	+14.9* [88/6]	+15.0* [86/3]	+13.6* [86/2]	+8.9* [64/1]	+4.8* [38/1]
tw1	FCN v2-small	+6.3* [15/2]	+1.6 [7/3]	+0.5 [1/0]	-1.3 [0/0]	-5.2* [0/0]
tw1	GraphCast Op.	+4.8* [17/2]	+7.2* [11/0]	+6.9* [5/0]	+4.6* [2/0]	+3.2 [0/0]
tw1	Pangu-Weather	+9.7* [21/1]	+10.7* [17/1]	+10.0* [16/0]	+6.2* [1/0]	+2.1 [0/0]
tw2	FCN v2-small	+5.6* [0/0]	-0.9 [0/0]	-1.9 [0/0]	-5.1 [0/0]	-7.0* [0/0]
tw2	GraphCast Op.	+4.7* [0/0]	+7.1* [0/0]	+5.8* [0/0]	+2.0 [0/0]	+3.7 [0/0]
tw2	Pangu-Weather	+9.3* [0/0]	+10.3* [1/0]	+8.1* [1/0]	+4.0 [0/0]	+0.9 [0/0]
oos_idr	FCN v2-small	+6.3* [36/6]	+6.9* [37/5]	+6.4* [33/2]	+4.6* [24/1]	+1.8 [12/1]
oos_idr	GraphCast Op.	+6.3* [46/4]	+9.3* [54/0]	+8.7* [45/0]	+7.9* [38/0]	+5.9* [26/0]
oos_idr	Pangu-Weather	+10.1* [61/2]	+11.1* [58/1]	+10.5* [58/0]	+6.7* [37/1]	+2.9* [13/1]
oos_tw1	FCN v2-small	+5.2* [0/0]	+3.1 [0/0]	+1.3 [0/0]	+1.8 [0/0]	-3.1 [0/0]
oos_tw1	GraphCast Op.	+5.6* [0/0]	+7.8* [0/0]	+7.3* [0/0]	+6.1* [0/0]	+1.4 [0/0]
oos_tw1	Pangu-Weather	+7.5* [1/0]	+10.6* [0/0]	+9.7* [0/0]	+8.8* [0/0]	+2.0 [0/0]
oos_tw2	FCN v2-small	+4.2 [0/0]	+1.6 [0/0]	-1.2 [0/0]	-2.4 [0/0]	-3.4 [0/0]
oos_tw2	GraphCast Op.	+6.2* [0/0]	+8.3* [0/0]	+7.0 [0/0]	+3.7 [0/0]	-0.2 [0/0]
oos_tw2	Pangu-Weather	+7.4* [0/0]	+11.0* [0/0]	+8.7* [0/0]	+7.3* [0/0]	+2.5 [0/0]
oos_emos	FCN v2-small	+5.0* [38/4]	+4.9* [32/3]	+4.0* [22/2]	+2.2* [10/0]	+0.0 [5/1]
oos_emos	GraphCast Op.	+5.6* [44/5]	+8.0* [48/1]	+7.0* [43/0]	+5.5* [31/0]	+3.3* [17/1]
oos_emos	Pangu-Weather	+9.2* [55/2]	+10.2* [62/0]	+9.3* [58/1]	+5.5* [36/1]	+2.6* [9/2]
oos_emos_tw1	FCN v2-small	+4.8* [0/0]	+2.2 [0/0]	+0.1 [0/0]	+0.4 [0/0]	-6.1* [0/0]
oos_emos_tw1	GraphCast Op.	+4.6* [0/0]	+7.8* [0/0]	+7.5* [0/0]	+6.2* [0/0]	+2.2 [0/0]
oos_emos_tw1	Pangu-Weather	+6.8* [0/0]	+9.7* [0/0]	+7.9* [0/0]	+5.2* [0/0]	-0.4 [0/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_core.csv; variable key: tmin.

Table S4. 10-m wind speed

Comparison set: **core**. Skill relative to GFS (%) at the selected lead times.

GraphCast Op. = GraphCast Operational; FCN v2-small = FourCastNet v2-small. All AI runs in this set use GFS initialization.

Score	Model	24 h	72 h	120 h	168 h	240 h
pcrps	FCN v2-small	-6.4* [5/60]	-3.5* [7/41]	-1.8* [6/24]	-0.3 [6/11]	+0.0 [3/3]
pcrps	GraphCast Op.	+0.6* [19/15]	+1.8* [22/8]	+1.8* [15/3]	+1.5* [11/4]	+0.8* [6/1]
pcrps	Pangu-Weather	-1.7* [11/29]	+0.9* [17/16]	+1.3* [14/7]	+0.9* [11/5]	+0.1 [1/2]
tw1	FCN v2-small	-7.4* [1/17]	-3.0* [2/6]	-1.2* [1/0]	+0.4 [1/0]	+0.4 [0/0]
tw1	GraphCast Op.	+1.0* [2/2]	+3.0* [2/0]	+2.4* [2/0]	+2.2* [2/0]	+1.2* [0/0]
tw1	Pangu-Weather	-2.4* [2/4]	+1.5* [2/1]	+1.6* [1/0]	+1.7* [1/0]	+0.6 [0/0]
tw2	FCN v2-small	-7.4* [1/4]	-2.7* [0/0]	-1.2 [0/0]	+0.4 [0/0]	+0.5 [0/0]
tw2	GraphCast Op.	+0.7* [2/0]	+3.0* [2/0]	+2.4* [1/0]	+2.3* [1/0]	+1.4* [0/0]
tw2	Pangu-Weather	-2.5* [2/2]	+1.4* [2/0]	+1.4* [0/0]	+1.8* [1/0]	+0.9* [0/0]
oos_idr	FCN v2-small	-5.3* [3/32]	-2.8* [3/16]	-1.2* [2/4]	-0.3 [1/1]	+0.0 [0/0]
oos_idr	GraphCast Op.	+0.9* [8/6]	+1.8* [8/3]	+1.6* [7/2]	+1.0* [3/2]	+0.6 [1/0]
oos_idr	Pangu-Weather	-1.1* [4/12]	+1.2* [7/5]	+1.6* [4/1]	+0.6 [1/1]	-0.2 [1/1]
oos_tw1	FCN v2-small	-4.3* [1/3]	-1.3* [0/0]	+0.3 [0/0]	-0.1 [0/0]	-0.1 [0/0]
oos_tw1	GraphCast Op.	+1.9* [1/0]	+3.2* [0/0]	+2.5* [1/0]	+1.3* [0/0]	+0.2 [0/0]
oos_tw1	Pangu-Weather	-1.1* [1/1]	+2.3* [1/0]	+2.6* [0/0]	+0.5 [0/0]	-0.3 [0/0]
oos_tw2	FCN v2-small	-3.3* [0/0]	-0.4 [0/0]	+0.7 [0/0]	-0.2 [0/0]	+0.0 [1/0]
oos_tw2	GraphCast Op.	+2.0* [1/0]	+3.4* [0/0]	+2.6* [0/0]	+1.0 [0/0]	+0.3 [2/1]
oos_tw2	Pangu-Weather	-0.7 [0/0]	+2.6* [1/0]	+2.6* [0/0]	+0.1 [0/0]	-0.3 [1/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_core.csv; variable key: ws.

Table S5. 24-h precipitation

Comparison set: **core**. Skill relative to GFS (%) at the selected lead times.

GraphCast Op. = GraphCast Operational; FCN v2-small = FourCastNet v2-small. All AI runs in this set use GFS initialization.

Score	Model	Day 1	Day 3	Day 5	Day 7	Day 10
pcrps	GraphCast Op.	+4.7* [8/1]	+6.5* [14/0]	+6.2* [14/0]	+5.1* [4/0]	+2.6* [0/0]
tw1	GraphCast Op.	+1.5* [0/0]	+3.7* [0/0]	+2.7* [0/0]	+2.7* [0/0]	+1.0* [0/0]
tw2	GraphCast Op.	+1.7* [0/0]	+4.1* [0/0]	+3.3* [0/0]	+3.1* [0/0]	+1.2* [0/0]
oos_idr	GraphCast Op.	+4.3* [0/0]	+3.8* [0/0]	+3.6* [0/0]	+2.1* [0/0]	+1.0 [1/0]
oos_tw1	GraphCast Op.	+1.8* [0/0]	+1.1 [0/0]	+0.4 [0/2]	+0.1 [0/0]	+1.2 [3/3]
oos_tw2	GraphCast Op.	+1.8* [0/1]	+1.4 [0/1]	+0.9 [1/2]	+0.3 [0/0]	+1.5* [6/2]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_core.csv; variable key: pr24.

Table S6. 2-m temperature

Comparison set: **core21**. Skill relative to GFS (%) at the selected lead times.

GraphCast Op. = GraphCast Operational; FCN v2-small = FourCastNet v2-small. All AI runs in this set use GFS initialization.

Score	Model	24 h	72 h	120 h	168 h	240 h
pcrps	FCN v2-small	+6.2* [60/19]	+7.3* [64/11]	+7.6* [66/9]	+7.0* [60/4]	+4.9* [50/1]
pcrps	Pangu-Weather	+8.5* [74/13]	+12.7* [81/6]	+11.5* [81/3]	+8.0* [68/2]	+2.8* [27/2]
tw1	FCN v2-small	+6.5* [13/3]	+7.0* [10/1]	+8.1* [7/0]	+7.8* [2/0]	+5.4* [0/0]
tw1	Pangu-Weather	+8.6* [25/2]	+11.7* [28/1]	+11.4* [26/0]	+9.5* [16/0]	+5.8* [0/0]
tw2	FCN v2-small	+4.6* [8/5]	+0.6 [3/2]	-1.1 [0/0]	-2.3 [0/0]	-3.3 [0/0]
tw2	Pangu-Weather	+7.4* [25/3]	+10.1* [12/2]	+7.9* [0/0]	+2.8* [0/0]	-0.7 [0/0]
oos_idr	FCN v2-small	+5.4* [45/13]	+6.5* [40/9]	+6.6* [36/4]	+6.6* [29/1]	+3.4* [17/1]
oos_idr	Pangu-Weather	+8.3* [61/7]	+12.5* [72/4]	+11.7* [66/2]	+8.5* [53/1]	+3.3* [6/0]
oos_tw1	FCN v2-small	+6.5* [0/0]	+5.9* [0/0]	+7.0* [0/0]	+8.2* [0/0]	+5.3* [0/0]
oos_tw1	Pangu-Weather	+7.6* [0/0]	+10.6* [0/0]	+10.2* [0/0]	+9.4* [0/0]	+4.0* [0/0]
oos_tw2	FCN v2-small	+5.5* [0/0]	+2.0 [0/0]	-0.1 [0/0]	+0.2 [0/0]	-1.8 [0/0]
oos_tw2	Pangu-Weather	+8.7* [0/0]	+11.5* [0/0]	+10.8* [0/0]	+8.7* [0/0]	+1.3 [0/0]
oos_emos	FCN v2-small	+3.4* [39/15]	+4.3* [34/8]	+4.2* [25/4]	+4.2* [23/1]	+1.5 [3/0]
oos_emos	Pangu-Weather	+7.3* [64/5]	+11.5* [72/3]	+10.7* [70/2]	+7.7* [63/0]	+3.9* [22/1]
oos_emos_tw1	FCN v2-small	+7.7* [0/0]	+6.4* [0/0]	+8.3* [0/0]	+10.2* [0/0]	+6.6* [0/0]
oos_emos_tw1	Pangu-Weather	+6.9* [0/0]	+10.0* [3/0]	+12.2* [0/0]	+12.4* [0/0]	+7.7* [0/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_core21.csv; variable key: t2.

Table S7. Daily maximum temperature

Comparison set: **core21**. Skill relative to GFS (%) at the selected lead times.

GraphCast Op. = GraphCast Operational; FCN v2-small = FourCastNet v2-small. All AI runs in this set use GFS initialization.

Score	Model	Day 1	Day 3	Day 5	Day 7	Day 9
pcrps	FCN v2-small	+2.8* [37/26]	+5.5* [37/10]	+6.1* [44/4]	+4.9* [42/4]	+4.2* [39/4]
pcrps	Pangu-Weather	+6.1* [46/19]	+8.4* [50/12]	+7.2* [48/11]	+2.5* [41/26]	+0.1 [29/26]
tw1	FCN v2-small	+1.2 [16/9]	+2.9* [3/0]	+5.3* [7/0]	+4.4* [6/0]	+3.8* [2/0]
tw1	Pangu-Weather	+3.9* [26/9]	+5.3* [16/1]	+6.3* [19/0]	+5.1* [12/0]	+3.8* [1/0]
tw2	FCN v2-small	-2.0 [4/5]	-0.4 [1/0]	+2.5 [1/0]	+2.3 [1/0]	+2.1 [0/0]
tw2	Pangu-Weather	+1.3 [11/5]	+2.2 [2/0]	+3.4* [2/0]	+3.2 [3/0]	+2.7* [0/0]
oos_idr	FCN v2-small	+4.0* [22/9]	+6.3* [28/5]	+5.0* [26/3]	+4.0* [20/3]	+1.5 [18/6]
oos_idr	Pangu-Weather	+6.0* [34/11]	+8.7* [38/5]	+6.8* [34/3]	+3.5* [29/9]	+0.5 [15/6]
oos_tw1	FCN v2-small	+3.7 [2/0]	+4.5* [1/0]	+6.0* [1/0]	+6.7* [2/0]	+6.1* [1/0]
oos_tw1	Pangu-Weather	+4.5* [2/0]	+5.9* [1/0]	+6.8* [1/0]	+6.8* [2/0]	+5.1* [0/0]
oos_tw2	FCN v2-small	+1.0 [0/0]	+1.2 [0/0]	+3.6 [0/0]	+4.8* [0/0]	+5.0* [0/0]
oos_tw2	Pangu-Weather	+2.2 [0/0]	+3.4* [0/0]	+4.7* [0/0]	+4.8* [0/0]	+3.4* [0/0]
oos_emos	FCN v2-small	+3.1* [10/4]	+5.3* [16/1]	+4.5* [10/0]	+3.6* [7/1]	+1.8 [2/0]
oos_emos	Pangu-Weather	+7.8* [47/4]	+10.2* [54/2]	+9.0* [46/2]	+6.1* [38/1]	+4.0* [18/1]
oos_emos_tw1	FCN v2-small	+9.7* [2/0]	+10.7* [2/0]	+12.6* [8/0]	+11.7* [0/0]	+9.9* [0/0]
oos_emos_tw1	Pangu-Weather	+11.3* [17/1]	+12.4* [3/0]	+13.2* [17/0]	+11.1* [1/0]	+8.8* [0/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_core21.csv; variable key: tmax.

Table S8. Daily minimum temperature

Comparison set: **core21**. Skill relative to GFS (%) at the selected lead times.

GraphCast Op. = GraphCast Operational; FCN v2-small = FourCastNet v2-small. All AI runs in this set use GFS initialization.

Score	Model	Day 1	Day 3	Day 5	Day 7	Day 9
pcrps	FCN v2-small	+12.0* [79/8]	+11.1* [78/8]	+9.7* [72/4]	+7.3* [61/4]	+4.6* [49/2]
pcrps	Pangu-Weather	+15.7* [89/6]	+15.7* [88/2]	+13.6* [89/2]	+8.4* [66/2]	+4.3* [41/2]
tw1	FCN v2-small	+8.1* [21/3]	+3.8* [11/3]	+2.6 [3/0]	+1.0 [1/0]	-2.1 [0/0]
tw1	Pangu-Weather	+11.1* [38/2]	+11.8* [34/1]	+10.4* [26/1]	+5.2* [2/0]	+1.4 [0/0]
tw2	FCN v2-small	+8.1* [0/0]	+2.3 [0/0]	+0.7 [0/0]	-1.8 [0/0]	-2.7 [0/0]
tw2	Pangu-Weather	+11.2* [12/0]	+11.5* [2/0]	+8.5* [1/0]	+2.9 [0/0]	+0.8 [0/0]
oos_idr	FCN v2-small	+6.6* [38/5]	+7.0* [38/7]	+6.5* [33/3]	+4.7* [27/1]	+1.7 [12/2]
oos_idr	Pangu-Weather	+10.3* [59/2]	+11.0* [58/1]	+10.4* [59/0]	+6.5* [41/1]	+2.6 [13/1]
oos_tw1	FCN v2-small	+5.7* [0/0]	+3.6 [0/0]	+1.7 [0/0]	+2.5 [0/0]	-4.0 [0/0]
oos_tw1	Pangu-Weather	+7.3* [1/0]	+10.3* [0/0]	+9.9* [0/0]	+8.4* [0/0]	+1.4 [0/0]
oos_tw2	FCN v2-small	+4.7 [0/0]	+1.7 [0/0]	-0.6 [0/0]	-0.5 [0/0]	-4.5 [0/0]
oos_tw2	Pangu-Weather	+6.8* [0/0]	+10.6* [0/0]	+8.6* [0/0]	+7.4* [0/0]	+1.2 [0/0]
oos_emos	FCN v2-small	+5.1* [36/3]	+5.0* [30/2]	+4.1* [22/2]	+2.3* [9/0]	+0.1 [5/1]
oos_emos	Pangu-Weather	+9.1* [56/2]	+9.9* [58/1]	+8.9* [56/1]	+5.1* [34/1]	+2.3* [8/2]
oos_emos_tw1	FCN v2-small	+5.4* [0/0]	+2.6 [0/0]	+0.4 [0/0]	+0.8 [0/0]	-5.1* [0/0]
oos_emos_tw1	Pangu-Weather	+7.3* [0/0]	+10.0* [0/0]	+8.2* [0/0]	+5.1* [0/0]	-0.4 [0/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_core21.csv; variable key: tmin.

Table S9. 10-m wind speed

Comparison set: **core21**. Skill relative to GFS (%) at the selected lead times.

GraphCast Op. = GraphCast Operational; FCN v2-small = FourCastNet v2-small. All AI runs in this set use GFS initialization.

Score	Model	24 h	72 h	120 h	168 h	240 h
pcrps	FCN v2-small	-6.4* [6/63]	-3.5* [7/44]	-1.8* [7/31]	-0.4 [7/16]	-0.1 [4/5]
pcrps	Pangu-Weather	-1.7* [12/35]	+0.9* [20/16]	+1.3* [18/10]	+0.7* [11/8]	+0.0 [3/4]
tw1	FCN v2-small	-7.3* [1/27]	-3.2* [2/9]	-1.3* [1/1]	+0.1 [1/0]	+0.2 [0/0]
tw1	Pangu-Weather	-2.4* [2/5]	+1.4* [4/3]	+1.5* [2/1]	+1.4* [2/0]	+0.4 [0/0]
tw2	FCN v2-small	-7.3* [1/8]	-3.0* [2/2]	-1.2* [1/0]	+0.1 [1/0]	+0.3 [0/0]
tw2	Pangu-Weather	-2.7* [1/2]	+1.2* [2/0]	+1.4* [1/0]	+1.4* [0/0]	+0.6 [0/0]
oos_idr	FCN v2-small	-5.5* [3/34]	-2.9* [3/18]	-1.3* [2/4]	-0.3 [1/1]	+0.2 [0/0]
oos_idr	Pangu-Weather	-1.4* [4/12]	+1.1* [6/5]	+1.5* [5/1]	+0.8 [1/1]	-0.0 [1/0]
oos_tw1	FCN v2-small	-5.1* [0/3]	-1.9* [0/0]	-0.1 [0/0]	+0.0 [0/0]	+0.3 [1/0]
oos_tw1	Pangu-Weather	-1.9* [1/2]	+1.7* [1/0]	+2.1* [0/0]	+1.0 [0/0]	-0.0 [0/0]
oos_tw2	FCN v2-small	-4.5* [0/0]	-1.4* [0/0]	+0.2 [0/0]	-0.0 [0/0]	+0.3 [2/0]
oos_tw2	Pangu-Weather	-2.0* [0/0]	+1.8* [1/0]	+1.9* [0/0]	+0.7 [0/0]	-0.1 [1/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_core21.csv; variable key: ws.

Table S10. 2-m temperature

Comparison set: **ic**. Skill relative to GFS (%) at the selected lead times.

Model initialization is GFS except where the ic model label explicitly says IFS. The reference forecast remains GFS.

Score	Model	24 h	72 h	120 h	168 h	240 h
pcrps	FCN v2-small / GFS	+6.0* [55/20]	+7.0* [57/10]	+7.1* [54/7]	+6.9* [48/2]	+5.5* [43/1]
pcrps	FCN v2-small / IFS	+8.1* [62/16]	+8.4* [62/9]	+9.2* [64/5]	+8.7* [62/2]	+7.1* [58/1]
pcrps	GraphCast Op. / GFS	+8.4* [70/10]	+11.4* [78/4]	+11.1* [79/2]	+10.2* [81/1]	+8.9* [78/1]
pcrps	GraphCast Op. / IFS	+18.8* [90/2]	+20.4* [93/1]	+19.9* [96/0]	+17.7* [95/0]	+13.2* [92/1]
pcrps	Pangu-Weather / GFS	+8.5* [68/11]	+12.8* [79/5]	+11.8* [78/3]	+9.0* [66/2]	+4.0* [22/0]
pcrps	Pangu-Weather / IFS	+17.1* [81/5]	+19.7* [89/3]	+18.2* [93/2]	+14.3* [89/0]	+7.3* [54/0]
tw1	FCN v2-small / GFS	+6.6* [10/3]	+7.2* [6/1]	+8.3* [2/0]	+9.4* [1/0]	+6.4* [0/0]
tw1	FCN v2-small / IFS	+9.4* [12/2]	+8.4* [7/1]	+9.8* [3/0]	+10.4* [2/0]	+6.9* [0/0]
tw1	GraphCast Op. / GFS	+9.6* [18/1]	+9.2* [3/0]	+9.3* [2/0]	+8.3* [0/0]	+8.2* [0/0]
tw1	GraphCast Op. / IFS	+18.2* [46/0]	+18.4* [44/0]	+17.5* [44/0]	+15.6* [36/0]	+12.8* [19/0]
tw1	Pangu-Weather / GFS	+8.6* [16/2]	+11.8* [22/1]	+11.5* [11/0]	+10.8* [10/0]	+6.9* [0/0]
tw1	Pangu-Weather / IFS	+15.9* [34/1]	+16.4* [34/0]	+15.4* [28/0]	+14.1* [19/0]	+8.8* [0/0]
tw2	FCN v2-small / GFS	+3.4* [0/0]	-2.0 [0/0]	-4.4 [0/0]	-7.1* [0/0]	-7.4* [0/0]
tw2	FCN v2-small / IFS	+5.9* [0/0]	-1.6 [0/0]	-3.7 [0/0]	-5.4 [0/0]	-9.1* [0/0]
tw2	GraphCast Op. / GFS	+5.1* [1/0]	+6.3* [0/0]	+4.3* [0/0]	+0.8 [0/0]	+3.0 [0/0]
tw2	GraphCast Op. / IFS	+15.6* [14/0]	+14.8* [0/0]	+13.4* [0/0]	+11.8* [0/0]	+3.6 [0/0]
tw2	Pangu-Weather / GFS	+6.9* [0/0]	+10.8* [0/0]	+7.5* [0/0]	+3.6 [0/0]	+1.2 [0/0]
tw2	Pangu-Weather / IFS	+15.4* [16/2]	+16.1* [0/0]	+12.5* [0/0]	+3.3 [0/0]	-3.6 [0/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_ic.csv; variable key: t2.

Table S11. Daily maximum temperature

Comparison set: **ic**. Skill relative to GFS (%) at the selected lead times.

Model initialization is GFS except where the ic model label explicitly says IFS. The reference forecast remains GFS.

Score	Model	Day 1	Day 3	Day 5	Day 7	Day 9
pcrps	FCN v2-small / GFS	+2.3* [30/24]	+4.5* [31/10]	+5.2* [32/4]	+4.6* [33/2]	+4.3* [31/3]
pcrps	FCN v2-small / IFS	+4.9* [34/17]	+6.9* [42/6]	+7.9* [42/2]	+7.0* [43/2]	+6.1* [39/1]
pcrps	GraphCast Op. / GFS	+7.9* [62/17]	+9.1* [62/9]	+8.8* [63/5]	+8.2* [59/3]	+7.1* [58/1]
pcrps	GraphCast Op. / IFS	+19.0* [88/1]	+20.5* [96/0]	+19.7* [96/0]	+16.9* [93/1]	+13.4* [91/1]
pcrps	Pangu-Weather / GFS	+6.4* [43/17]	+8.5* [47/8]	+7.8* [50/4]	+3.5* [39/14]	+0.6 [19/11]
pcrps	Pangu-Weather / IFS	+13.5* [71/3]	+14.7* [78/2]	+13.3* [77/1]	+9.2* [54/1]	+4.4* [32/2]
tw1	FCN v2-small / GFS	-0.4 [6/3]	+1.7 [1/0]	+4.9* [3/0]	+4.9* [2/0]	+4.0* [1/0]
tw1	FCN v2-small / IFS	+2.0 [9/2]	+3.7* [5/0]	+6.2* [1/0]	+7.1* [3/0]	+5.3* [2/0]
tw1	GraphCast Op. / GFS	+1.5 [13/7]	+2.3 [2/1]	+4.9* [1/0]	+5.6* [0/0]	+7.1* [2/0]
tw1	GraphCast Op. / IFS	+14.6* [51/0]	+15.5* [54/0]	+15.4* [48/0]	+14.7* [51/0]	+12.7* [35/1]
tw1	Pangu-Weather / GFS	+2.6 [12/5]	+4.6* [9/1]	+5.5* [6/0]	+5.7* [9/0]	+4.6* [0/0]
tw1	Pangu-Weather / IFS	+10.6* [27/0]	+10.8* [31/0]	+11.0* [23/0]	+10.1* [17/0]	+7.4* [9/0]
tw2	FCN v2-small / GFS	-4.0* [2/2]	-1.8 [0/0]	+2.2 [0/0]	+2.7 [0/0]	+2.4 [0/0]
tw2	FCN v2-small / IFS	-1.4 [1/1]	+0.1 [0/0]	+3.1* [0/0]	+5.0* [0/0]	+3.7* [0/0]
tw2	GraphCast Op. / GFS	-1.2 [1/2]	-0.8 [0/0]	+2.1 [0/0]	+3.2* [0/0]	+5.6* [0/0]
tw2	GraphCast Op. / IFS	+12.5* [22/1]	+13.3* [21/0]	+13.6* [21/0]	+13.3* [21/0]	+11.6* [6/0]
tw2	Pangu-Weather / GFS	-0.3 [1/2]	+1.2 [1/0]	+2.3 [0/0]	+3.5 [1/0]	+3.3* [0/0]
tw2	Pangu-Weather / IFS	+8.4* [6/0]	+7.9* [4/0]	+8.2* [0/0]	+8.0* [2/0]	+5.8* [0/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_ic.csv; variable key: tmax.

Table S12. Daily minimum temperature

Comparison set: **ic**. Skill relative to GFS (%) at the selected lead times.

Model initialization is GFS except where the ic model label explicitly says IFS. The reference forecast remains GFS.

Score	Model	Day 1	Day 3	Day 5	Day 7	Day 9
pcrps	FCN v2-small / GFS	+11.5* [76/8]	+10.6* [72/7]	+9.1* [59/4]	+6.7* [47/3]	+4.9* [41/2]
pcrps	FCN v2-small / IFS	+13.2* [78/5]	+11.6* [74/5]	+10.9* [73/3]	+8.6* [62/2]	+7.3* [54/1]
pcrps	GraphCast Op. / GFS	+9.4* [75/7]	+12.6* [83/0]	+12.1* [82/0]	+10.4* [77/0]	+9.1* [74/0]
pcrps	GraphCast Op. / IFS	+18.2* [93/1]	+19.0* [93/0]	+18.6* [97/0]	+15.8* [97/0]	+12.6* [93/1]
pcrps	Pangu-Weather / GFS	+15.5* [87/5]	+15.8* [87/3]	+14.2* [84/2]	+9.5* [64/0]	+5.3* [37/0]
pcrps	Pangu-Weather / IFS	+20.2* [92/3]	+20.4* [92/2]	+18.4* [93/1]	+13.3* [84/0]	+8.2* [57/0]
tw1	FCN v2-small / GFS	+7.0* [6/2]	+1.5 [0/0]	+0.1 [0/0]	-2.2 [0/0]	-6.3* [0/0]
tw1	FCN v2-small / IFS	+10.0* [13/3]	+2.7 [0/0]	+1.1 [1/0]	-0.6 [0/0]	-7.0* [0/0]
tw1	GraphCast Op. / GFS	+5.1* [12/0]	+7.6* [3/0]	+7.0* [0/0]	+4.8* [0/0]	+4.3 [0/0]
tw1	GraphCast Op. / IFS	+16.4* [36/0]	+15.3* [26/0]	+14.6* [13/0]	+13.6* [3/0]	+6.6* [0/0]
tw1	Pangu-Weather / GFS	+10.7* [21/0]	+11.7* [12/0]	+10.7* [12/0]	+6.8* [0/0]	+2.7 [0/0]
tw1	Pangu-Weather / IFS	+18.1* [28/1]	+17.2* [22/0]	+14.9* [13/0]	+8.0* [2/0]	+3.2 [1/0]
tw2	FCN v2-small / GFS	+6.3* [0/0]	-0.8 [0/0]	-2.5 [0/0]	-6.0 [0/0]	-8.1* [0/0]
tw2	FCN v2-small / IFS	+8.4* [0/0]	-0.3 [0/0]	-2.2 [0/0]	-4.5 [0/0]	-9.8* [0/0]
tw2	GraphCast Op. / GFS	+4.7* [0/0]	+7.9* [0/0]	+6.0* [0/0]	+2.0 [0/0]	+5.6 [0/0]
tw2	GraphCast Op. / IFS	+15.8* [0/0]	+14.9* [0/0]	+14.0* [0/0]	+12.3* [0/0]	+4.3 [0/0]
tw2	Pangu-Weather / GFS	+9.9* [0/0]	+11.4* [0/0]	+8.5* [0/0]	+3.9 [0/0]	+1.1 [0/0]
tw2	Pangu-Weather / IFS	+16.8* [0/0]	+15.6* [0/0]	+12.7* [0/0]	+2.9 [0/0]	-1.1 [0/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_ic.csv; variable key: tmin.

Table S13. 10-m wind speed

Comparison set: **ic**. Skill relative to GFS (%) at the selected lead times.

Model initialization is GFS except where the ic model label explicitly says IFS. The reference forecast remains GFS.

Score	Model	24 h	72 h	120 h	168 h	240 h
pcrps	FCN v2-small / GFS	-6.4* [4/57]	-3.5* [6/36]	-1.7* [6/21]	-0.2 [5/8]	+0.1 [2/1]
pcrps	FCN v2-small / IFS	-5.7* [5/51]	-2.5* [7/28]	-0.4 [7/14]	+0.3 [6/6]	+0.4 [3/2]
pcrps	GraphCast Op. / GFS	+0.7* [18/11]	+1.9* [21/7]	+1.8* [10/3]	+1.5* [10/3]	+0.8* [5/1]
pcrps	GraphCast Op. / IFS	+3.1* [32/7]	+5.6* [52/2]	+5.9* [55/1]	+4.7* [41/1]	+2.2* [20/0]
pcrps	Pangu-Weather / GFS	-1.8* [7/25]	+1.0* [17/14]	+1.4* [13/6]	+0.9* [9/4]	+0.2 [0/1]
pcrps	Pangu-Weather / IFS	-0.7* [13/19]	+2.8* [30/8]	+3.4* [27/3]	+2.4* [19/2]	+0.7* [0/0]
tw1	FCN v2-small / GFS	-7.3* [1/17]	-2.9* [2/4]	-1.2 [1/0]	+0.4 [1/0]	+0.5 [0/0]
tw1	FCN v2-small / IFS	-6.3* [1/12]	-1.8* [1/2]	+0.7 [1/0]	+1.0 [1/0]	+0.9* [0/0]
tw1	GraphCast Op. / GFS	+1.1* [3/2]	+2.9* [2/0]	+2.5* [2/0]	+2.3* [2/0]	+1.3* [0/0]
tw1	GraphCast Op. / IFS	+4.0* [7/1]	+7.0* [11/1]	+6.7* [10/0]	+5.9* [5/0]	+2.9* [1/0]
tw1	Pangu-Weather / GFS	-2.4* [2/4]	+1.6* [2/0]	+1.6* [2/0]	+1.7* [1/0]	+0.7* [0/0]
tw1	Pangu-Weather / IFS	-0.8* [2/3]	+4.0* [4/0]	+4.5* [2/0]	+2.8* [1/0]	+1.2* [0/0]
tw2	FCN v2-small / GFS	-7.3* [1/4]	-2.5* [0/0]	-1.3 [0/0]	+0.4 [0/0]	+0.7 [0/0]
tw2	FCN v2-small / IFS	-6.2* [1/4]	-1.5* [0/0]	+0.7 [0/0]	+0.9 [0/0]	+1.3* [0/0]
tw2	GraphCast Op. / GFS	+0.8* [2/0]	+2.9* [1/0]	+2.4* [1/0]	+2.5* [1/0]	+1.5* [0/0]
tw2	GraphCast Op. / IFS	+4.1* [4/0]	+7.0* [5/0]	+6.3* [2/0]	+5.9* [2/0]	+3.0* [0/0]
tw2	Pangu-Weather / GFS	-2.6* [1/0]	+1.4* [1/0]	+1.3* [0/0]	+1.7* [0/0]	+1.0* [0/0]
tw2	Pangu-Weather / IFS	-0.9* [1/0]	+4.0* [1/0]	+4.4* [1/0]	+2.6* [0/0]	+1.3* [0/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_ic.csv; variable key: ws.

Table S14. 24-h precipitation

Comparison set: **ic**. Skill relative to GFS (%) at the selected lead times.

Model initialization is GFS except where the ic model label explicitly says IFS. The reference forecast remains GFS.

Score	Model	Day 1	Day 3	Day 5	Day 7	Day 10
pcrps	GraphCast Op. / GFS	+4.8* [8/1]	+6.6* [12/0]	+6.2* [12/0]	+5.5* [3/0]	+2.7* [0/0]
pcrps	GraphCast Op. / IFS	+9.4* [57/0]	+12.0* [65/0]	+11.7* [63/0]	+9.6* [52/0]	+5.8* [35/0]
tw1	GraphCast Op. / GFS	+1.6* [0/0]	+3.7* [0/0]	+2.7* [0/0]	+3.1* [0/0]	+1.1* [0/0]
tw1	GraphCast Op. / IFS	+5.1* [0/0]	+6.6* [0/0]	+5.9* [0/0]	+4.9* [0/0]	+2.5* [0/0]
tw2	GraphCast Op. / GFS	+1.7* [0/0]	+4.1* [0/0]	+3.3* [0/0]	+3.5* [0/0]	+1.4* [0/0]
tw2	GraphCast Op. / IFS	+5.2* [0/0]	+6.9* [0/0]	+6.5* [0/0]	+5.4* [0/0]	+2.8* [0/0]

*: 95% calendar-block bootstrap interval excludes zero.
[better/worse]: percentage of stations significant at BH-FDR 5%. Positive skill favours the AI model. Score keys and limitations are defined on page 2.

Source: results/tables/key_results_ic.csv; variable key: pr24.

Table S15. Station inventory (1/9)

Analysis station IDs, coordinates, climate groups and daily-variable eligibility. Rows 1-40 of 334.

All 334 stations are used for 2-m temperature and wind speed. Tmax: 180; Tmin: 180; P24 (24-h precipitation): 249. Y = eligible; - = not eligible.

Station ID	Latitude	Longitude	Climate group	Tmax	Tmin	P24
50136099999	52.967	122.533	Cold	-	-	-
50353099999	51.733	126.633	Cold	Y	Y	Y
50434099999	50.45	121.7	Cold	Y	Y	-
50468099999	50.25	127.45	Cold	-	-	-
50548099999	49.2	123.717	Cold	-	-	-
50557099999	49.167	125.233	Cold	Y	Y	Y
50564099999	49.433	127.35	Cold	Y	Y	Y
50603099999	48.683	116.817	Arid	-	-	-
50632099999	48.767	121.917	Cold	Y	Y	Y
50658099999	48.05	125.883	Cold	Y	Y	Y
50727099999	47.167	119.933	Cold	Y	Y	Y
50745099999	47.24	123.918	Cold	Y	Y	Y
50756099999	47.45	126.867	Cold	Y	Y	Y
50774099999	47.7	128.833	Cold	-	-	Y
50788099999	47.233	131.983	Cold	Y	Y	Y
50854099999	46.383	125.317	Cold	Y	Y	Y
50888099999	46.317	132.183	Cold	-	-	Y
50915099999	45.517	116.967	Arid	Y	Y	Y
50949099999	45.083	124.867	Cold	Y	Y	Y
50953099999	45.933	126.567	Cold	Y	Y	Y
50963099999	45.967	128.733	Cold	Y	Y	Y
50968099999	45.217	127.967	Cold	Y	Y	Y
50978099999	45.3	130.917	Cold	Y	Y	Y
50983099999	45.767	132.967	Cold	-	-	Y
51053099999	48.05	86.35	Arid	-	-	-
51076099999	47.733	88.083	Arid	Y	Y	Y
51087099999	46.983	89.517	Arid	Y	Y	Y
51133099999	46.733	83.0	Cold	-	-	Y
51156099999	46.817	85.75	Arid	Y	Y	Y
51243099999	45.617	84.85	Arid	Y	Y	Y
51334099999	44.567	82.817	Arid	Y	Y	Y
51379099999	44.017	89.567	Arid	Y	Y	Y
51463099999	43.783	87.617	Arid	Y	Y	Y
51495099999	43.217	91.733	Arid	-	-	-
51542099999	43.033	84.15	Cold	-	-	Y
51573099999	42.95	89.23	Arid	Y	Y	Y
51644099999	41.717	82.95	Arid	Y	Y	Y
51656099999	41.733	85.817	Arid	Y	Y	-
51711099999	40.933	78.45	Arid	-	-	-
51716099999	39.8	78.567	Arid	Y	Y	Y

Source: data/interim/stations_analysis.csv. Coordinates are degrees north/east. Full station names, elevation, climate codes and flags are retained in the public CSV.

Table S15. Station inventory (2/9)

Analysis station IDs, coordinates, climate groups and daily-variable eligibility. Rows 41-80 of 334.

All 334 stations are used for 2-m temperature and wind speed. Tmax: 180; Tmin: 180; P24 (24-h precipitation): 249. Y = eligible; - = not eligible.

Station ID	Latitude	Longitude	Climate group	Tmax	Tmin	P24
51730099999	40.5	81.05	Arid	-	-	-
51747099999	39.0	83.667	Arid	Y	Y	-
51765099999	40.633	87.7	Arid	Y	Y	-
51811099999	38.433	77.267	Arid	Y	Y	Y
51818099999	37.617	78.283	Arid	-	-	-
51828099999	37.133	79.933	Arid	Y	Y	Y
52112099999	43.6	95.133	Arid	-	-	-
52203099999	42.817	93.517	Arid	Y	Y	-
52323099999	41.8	97.033	Arid	Y	Y	-
52378099999	41.367	102.367	Arid	-	-	-
52436099999	40.267	97.033	Arid	Y	Y	Y
52495099999	40.75	104.5	Arid	Y	Y	Y
52533099999	39.767	98.483	Arid	Y	Y	Y
52602099999	38.833	93.383	Arid	Y	Y	-
52652099999	39.083	100.283	Arid	-	-	-
52681099999	38.633	103.083	Arid	Y	Y	Y
52713099999	37.85	95.367	Highland	Y	Y	Y
52737099999	37.367	97.367	Arid	-	-	-
52754099999	37.333	100.133	Highland	Y	Y	Y
52787099999	37.2	102.867	Highland	Y	Y	Y
52818099999	36.417	94.9	Arid	Y	Y	Y
52908099999	35.217	93.083	Highland	-	-	Y
52955099999	35.583	100.75	Highland	Y	Y	Y
52983099999	35.867	104.15	Arid	Y	Y	Y
52996099999	35.383	105.0	Cold	-	-	Y
53083099999	44.617	114.15	Arid	-	-	-
53149099999	42.533	110.133	Arid	-	-	-
53192099999	44.017	115.0	Arid	Y	Y	Y
53231099999	41.45	106.383	Arid	Y	Y	-
53336099999	41.567	108.517	Arid	Y	Y	Y
53352099999	41.7	110.433	Arid	Y	Y	Y
53391099999	41.9	114.0	Arid	Y	Y	Y
53463099999	40.851	111.824	Arid	Y	Y	Y
53480099999	41.033	113.067	Arid	-	-	-
53487099999	40.083	113.417	Arid	Y	Y	Y
53502099999	39.783	105.75	Arid	Y	Y	Y
53513099999	40.767	107.4	Arid	-	-	-
53529099999	39.1	107.983	Arid	Y	Y	Y
53543099999	39.833	109.983	Arid	-	-	-
53564099999	39.367	111.217	Cold	-	-	-

Source: data/interim/stations_analysis.csv. Coordinates are degrees north/east. Full station names, elevation, climate codes and flags are retained in the public CSV.

Table S15. Station inventory (3/9)

Analysis station IDs, coordinates, climate groups and daily-variable eligibility. Rows 81-120 of 334.

All 334 stations are used for 2-m temperature and wind speed. Tmax: 180; Tmin: 180; P24 (24-h precipitation): 249. Y = eligible; - = not eligible.

Station ID	Latitude	Longitude	Climate group	Tmax	Tmin	P24
53588099999	38.95	113.517	Cold	-	-	-
53593099999	39.833	114.567	Arid	-	-	-
53646099999	38.267	109.783	Arid	Y	Y	Y
53698099999	38.067	114.35	Arid	Y	Y	Y
53705099999	37.483	105.683	Arid	-	-	-
53723099999	37.8	107.383	Arid	Y	Y	Y
53764099999	37.5	111.1	Cold	-	-	-
53772099999	37.747	112.628	Arid	Y	Y	Y
53787099999	37.067	112.983	Cold	-	-	-
53798099999	37.183	114.367	Cold	-	-	-
53845099999	36.6	109.5	Cold	Y	Y	Y
53863099999	37.033	111.917	Arid	Y	Y	Y
53898099999	36.05	114.133	Cold	Y	Y	Y
53915099999	35.55	106.667	Cold	Y	Y	Y
53923099999	35.733	107.633	Cold	-	-	-
53959099999	35.117	111.067	Arid	Y	Y	Y
53975099999	35.483	112.4	Cold	-	-	-
54012099999	44.583	117.6	Arid	Y	Y	Y
54026099999	44.567	120.9	Arid	Y	Y	Y
54027099999	43.983	119.4	Arid	Y	Y	Y
54049099999	44.25	123.967	Cold	-	-	-
54094099999	44.5	129.667	Cold	Y	Y	Y
54096099999	44.383	131.15	Cold	Y	Y	Y
54102099999	43.95	116.117	Arid	Y	Y	Y
54115099999	43.633	118.033	Arid	Y	Y	Y
54135099999	43.6	122.267	Arid	Y	Y	Y
54157099999	43.183	124.333	Cold	Y	Y	Y
54161099999	43.996	125.681	Cold	Y	Y	Y
54186099999	43.367	128.2	Cold	-	-	-
54208099999	42.183	116.467	Cold	Y	Y	Y
54218099999	42.3	118.833	Arid	Y	Y	Y
54226099999	42.333	120.7	Cold	-	-	-
54236099999	42.417	122.533	Cold	Y	Y	Y
54259099999	42.067	124.867	Cold	-	-	-
54273099999	42.983	126.75	Cold	-	-	Y
54292099999	42.883	129.5	Cold	Y	Y	Y
54308099999	41.2	116.633	Cold	-	-	-
54311099999	41.933	117.75	Cold	-	-	-
54324099999	41.55	120.45	Arid	Y	Y	Y
54337099999	41.133	121.117	Cold	Y	Y	Y

Source: data/interim/stations_analysis.csv. Coordinates are degrees north/east. Full station names, elevation, climate codes and flags are retained in the public CSV.

Table S15. Station inventory (4/9)

Analysis station IDs, coordinates, climate groups and daily-variable eligibility. Rows 121-160 of 334.

All 334 stations are used for 2-m temperature and wind speed. Tmax: 180; Tmin: 180; P24 (24-h precipitation): 249. Y = eligible; - = not eligible.

Station ID	Latitude	Longitude	Climate group	Tmax	Tmin	P24
54342099999	41.733	123.517	Cold	Y	Y	Y
54346099999	41.317	123.783	Cold	Y	Y	Y
54374099999	41.8	126.917	Cold	Y	Y	Y
54377099999	41.15	126.217	Cold	-	-	-
54386099999	41.417	128.183	Cold	-	-	Y
54401099999	40.783	114.883	Arid	-	-	-
54405099999	40.417	115.5	Arid	Y	Y	Y
54423099999	40.967	117.917	Cold	Y	Y	Y
54471099999	40.667	122.2	Cold	Y	Y	Y
54493099999	40.717	124.783	Cold	-	-	-
54497099999	40.05	124.333	Cold	Y	Y	Y
54511099999	40.08	116.585	Arid	Y	Y	Y
54527099999	39.1	117.167	Arid	Y	Y	Y
54539099999	39.433	118.9	Cold	Y	Y	Y
54602099999	38.733	115.483	Arid	-	-	-
54618099999	38.083	116.55	Arid	Y	Y	Y
54662099999	38.966	121.539	Cold	Y	Y	Y
54715099999	37.333	116.567	Arid	-	-	-
54725099999	37.5	117.533	Cold	Y	Y	Y
54751099999	37.933	120.717	Temperate	-	-	-
54753099999	37.617	120.317	Cold	-	-	-
54776099999	37.4	122.683	Temperate	Y	Y	Y
54808099999	36.233	115.667	Arid	-	-	-
54826099999	36.25	117.1	Cold	-	-	-
54836099999	36.183	118.15	Cold	-	-	-
54843099999	36.767	119.183	Cold	Y	Y	Y
54863099999	36.767	121.167	Cold	-	-	-
54909099999	35.083	115.517	Temperate	Y	Y	Y
54916099999	35.567	116.85	Cold	Y	Y	Y
54929099999	35.25	117.95	Cold	-	-	-
54945099999	35.433	119.533	Temperate	-	-	-
55228099999	32.5	80.083	Highland	Y	Y	-
55299099999	31.483	92.067	Highland	Y	Y	Y
55578099999	29.25	88.883	Highland	Y	Y	-
55664099999	28.633	87.083	Highland	-	-	-
55773099999	27.733	89.083	Highland	-	-	Y
56004099999	34.217	92.433	Highland	Y	Y	Y
56018099999	32.9	95.3	Highland	-	-	Y
56021099999	34.133	95.783	Highland	Y	Y	Y
56029099999	33.017	96.95	Highland	Y	Y	Y

Source: data/interim/stations_analysis.csv. Coordinates are degrees north/east. Full station names, elevation, climate codes and flags are retained in the public CSV.

Table S15. Station inventory (5/9)

Analysis station IDs, coordinates, climate groups and daily-variable eligibility. Rows 161-200 of 334.

All 334 stations are used for 2-m temperature and wind speed. Tmax: 180; Tmin: 180; P24 (24-h precipitation): 249. Y = eligible; - = not eligible.

Station ID	Latitude	Longitude	Climate group	Tmax	Tmin	P24
56033099999	34.917	98.217	Highland	Y	Y	Y
56046099999	33.75	99.65	Highland	Y	Y	Y
56065099999	34.733	101.6	Highland	-	-	Y
56079099999	33.583	102.967	Highland	Y	Y	Y
56096099999	33.4	104.917	Arid	Y	Y	Y
56116099999	31.417	95.6	Highland	Y	Y	Y
56137099999	31.15	97.167	Highland	Y	Y	Y
56144099999	31.733	98.567	Highland	-	-	Y
56146099999	31.617	100.0	Highland	Y	Y	Y
56152099999	32.283	100.333	Highland	-	-	Y
56167099999	30.983	101.117	Cold	-	-	Y
56172099999	31.9	102.233	Cold	Y	Y	Y
56182099999	32.667	103.6	Cold	Y	Y	Y
56187099999	30.75	103.867	Temperate	Y	Y	Y
56196099999	31.45	104.733	Temperate	-	-	Y
56257099999	30.0	100.267	Highland	Y	Y	Y
56287099999	29.983	103.0	Temperate	-	-	Y
56294099999	30.579	103.947	Temperate	-	-	-
56312099999	29.567	94.467	Highland	-	-	-
56357099999	29.05	100.3	Highland	-	-	Y
56374099999	30.05	101.967	Cold	-	-	Y
56385099999	29.517	103.333	Highland	-	-	Y
56444099999	28.45	98.883	Highland	Y	Y	Y
56492099999	28.8	104.6	Temperate	Y	Y	Y
56571099999	27.9	102.267	Temperate	Y	Y	Y
56586099999	27.333	103.75	Temperate	-	-	Y
56651099999	26.833	100.217	Temperate	Y	Y	Y
56671099999	26.65	102.25	Temperate	Y	Y	Y
56684099999	26.4	103.25	Temperate	-	-	Y
56691099999	26.867	104.283	Temperate	-	-	-
56739099999	24.983	98.5	Temperate	Y	Y	Y
56748099999	25.117	99.183	Temperate	-	-	Y
56751099999	25.7	100.183	Temperate	-	-	Y
56786099999	25.583	103.833	Temperate	-	-	Y
56886099999	24.533	103.767	Temperate	-	-	Y
56946099999	23.55	99.4	Temperate	-	-	Y
56951099999	23.95	100.217	Temperate	Y	Y	Y
56954099999	22.567	99.933	Temperate	Y	Y	Y
56959099999	22.0	100.783	Temperate	-	-	Y
56964099999	22.767	100.983	Temperate	Y	Y	Y

Source: data/interim/stations_analysis.csv. Coordinates are degrees north/east. Full station names, elevation, climate codes and flags are retained in the public CSV.

Table S15. Station inventory (6/9)

Analysis station IDs, coordinates, climate groups and daily-variable eligibility. Rows 201-240 of 334.

All 334 stations are used for 2-m temperature and wind speed. Tmax: 180; Tmin: 180; P24 (24-h precipitation): 249. Y = eligible; - = not eligible.

Station ID	Latitude	Longitude	Climate group	Tmax	Tmin	P24
56966099999	23.6	101.983	Temperate	-	-	Y
56969099999	21.5	101.583	Temperate	-	-	Y
56977099999	22.617	101.817	Temperate	-	-	Y
56985099999	23.45	103.333	Temperate	Y	Y	Y
57025099999	34.517	107.383	Temperate	-	-	-
57036099999	34.447	108.752	Arid	-	-	-
57046099999	34.483	110.083	Cold	-	-	-
57067099999	34.083	111.067	Cold	Y	Y	Y
57071099999	34.817	112.433	Temperate	-	-	-
57083099999	34.52	113.841	Temperate	Y	Y	Y
57127099999	33.067	107.033	Temperate	Y	Y	Y
57131099999	34.433	108.967	Arid	Y	Y	Y
57178099999	33.1	112.483	Temperate	-	-	-
57193099999	33.783	114.517	Temperate	-	-	-
57237099999	32.067	108.033	Temperate	Y	Y	Y
57245099999	32.717	109.033	Temperate	-	-	-
57259099999	32.033	110.767	Temperate	-	-	Y
57265099999	32.433	111.733	Temperate	Y	Y	Y
57279099999	32.15	112.667	Temperate	-	-	-
57290099999	32.933	113.917	Temperate	Y	Y	Y
57297099999	32.133	114.05	Temperate	Y	Y	Y
57306099999	31.583	105.967	Temperate	-	-	Y
57328099999	31.2	107.5	Temperate	-	-	-
57348099999	31.017	109.533	Temperate	-	-	-
57378099999	31.167	112.633	Temperate	-	-	-
57399099999	31.133	114.95	Temperate	-	-	Y
57411099999	30.75	106.133	Temperate	Y	Y	Y
57426099999	30.683	107.8	Temperate	-	-	Y
57447099999	30.283	109.467	Temperate	Y	Y	Y
57461099999	30.733	111.367	Temperate	Y	Y	Y
57476099999	30.333	112.183	Temperate	-	-	-
57494099999	30.784	114.208	Temperate	Y	Y	Y
57516099999	29.719	106.642	Temperate	Y	Y	Y
57554099999	29.4	110.167	Temperate	-	-	Y
57584099999	29.383	113.083	Temperate	-	-	Y
57598099999	29.033	114.583	Temperate	-	-	Y
57604099999	28.783	105.383	Temperate	-	-	Y
57633099999	28.833	108.767	Temperate	Y	Y	Y
57655099999	28.467	110.4	Temperate	-	-	Y
57662099999	29.05	111.683	Temperate	Y	Y	Y

Source: data/interim/stations_analysis.csv. Coordinates are degrees north/east. Full station names, elevation, climate codes and flags are retained in the public CSV.

Table S15. Station inventory (7/9)

Analysis station IDs, coordinates, climate groups and daily-variable eligibility. Rows 241-280 of 334.

All 334 stations are used for 2-m temperature and wind speed. Tmax: 180; Tmin: 180; P24 (24-h precipitation): 249. Y = eligible; - = not eligible.

Station ID	Latitude	Longitude	Climate group	Tmax	Tmin	P24
57687099999	28.117	112.783	Temperate	Y	Y	Y
57707099999	27.3	105.233	Temperate	Y	Y	Y
57731099999	27.95	108.25	Temperate	-	-	Y
57745099999	27.45	109.683	Temperate	Y	Y	Y
57766099999	27.183	111.45	Temperate	-	-	Y
57776099999	27.3	112.7	Temperate	-	-	Y
57793099999	27.8	114.383	Temperate	-	-	Y
57799099999	27.117	114.967	Temperate	Y	Y	Y
57816099999	26.539	106.801	Temperate	Y	Y	Y
57832099999	26.967	108.667	Temperate	-	-	Y
57845099999	26.167	109.783	Temperate	-	-	Y
57853099999	26.733	110.633	Temperate	-	-	Y
57866099999	26.233	111.617	Temperate	Y	Y	Y
57902099999	25.433	105.183	Temperate	Y	Y	Y
57916099999	25.433	106.767	Temperate	-	-	Y
57922099999	25.833	107.55	Temperate	-	-	Y
57932099999	25.967	108.533	Temperate	-	-	Y
57957099999	25.218	110.039	Temperate	Y	Y	Y
57972099999	25.733	112.967	Temperate	-	-	-
57993099999	25.85	114.95	Temperate	Y	Y	Y
58027099999	34.283	117.15	Temperate	Y	Y	Y
58040099999	34.85	119.133	Temperate	Y	Y	Y
58102099999	33.783	115.733	Temperate	Y	Y	Y
58141099999	33.633	118.933	Temperate	-	-	-
58203099999	32.867	115.733	Temperate	-	-	-
58208099999	32.167	115.617	Temperate	-	-	-
58238099999	31.742	118.862	Temperate	Y	Y	Y
58251099999	32.85	120.283	Temperate	Y	Y	Y
58265099999	32.067	121.6	Temperate	-	-	-
58314099999	31.4	116.333	Temperate	Y	Y	Y
58338099999	31.15	118.583	Temperate	-	-	Y
58345099999	31.433	119.483	Temperate	-	-	Y
58362099999	31.4	121.467	Temperate	Y	Y	Y
58367099999	31.198	121.336	Temperate	-	-	-
58424099999	30.617	116.967	Temperate	Y	Y	Y
58437099999	30.133	118.15	Temperate	-	-	Y
58457099999	30.228	120.432	Temperate	Y	Y	Y
58472099999	30.733	122.45	Temperate	-	-	Y
58477099999	30.033	122.117	Temperate	Y	Y	Y
58506099999	29.583	115.983	Temperate	-	-	Y

Source: data/interim/stations_analysis.csv. Coordinates are degrees north/east. Full station names, elevation, climate codes and flags are retained in the public CSV.

Table S15. Station inventory (8/9)

Analysis station IDs, coordinates, climate groups and daily-variable eligibility. Rows 281-320 of 334.

All 334 stations are used for 2-m temperature and wind speed. Tmax: 180; Tmin: 180; P24 (24-h precipitation): 249. Y = eligible; - = not eligible.

Station ID	Latitude	Longitude	Climate group	Tmax	Tmin	P24
58556099999	29.6	120.817	Temperate	-	-	Y
58606099999	28.865	115.9	Temperate	Y	Y	Y
58633099999	28.967	118.867	Temperate	Y	Y	Y
58646099999	28.45	119.917	Temperate	-	-	Y
58666099999	28.45	121.883	Temperate	Y	Y	Y
58715099999	27.583	116.65	Temperate	Y	Y	Y
58725099999	27.333	117.467	Temperate	-	-	Y
58730099999	27.767	118.033	Temperate	-	-	Y
58731099999	27.917	118.533	Temperate	-	-	Y
58752099999	27.783	120.65	Temperate	Y	Y	Y
58754099999	27.333	120.2	Temperate	-	-	Y
58813099999	26.85	116.333	Temperate	-	-	Y
58834099999	26.633	118.15	Temperate	Y	Y	Y
58847099999	26.083	119.283	Temperate	Y	Y	Y
58911099999	25.85	116.367	Temperate	-	-	Y
58921099999	25.95	117.37	Temperate	Y	Y	Y
58926099999	25.3	117.4	Temperate	-	-	Y
58931099999	25.717	118.1	Temperate	-	-	Y
58944099999	25.517	119.783	Temperate	-	-	Y
58968099999	25.033	121.517	Temperate	-	-	-
58974099999	25.633	122.067	Temperate	-	-	-
59007099999	24.067	105.067	Temperate	-	-	Y
59023099999	24.7	108.05	Temperate	Y	Y	Y
59058099999	24.2	110.517	Temperate	-	-	Y
59072099999	24.8	112.367	Temperate	-	-	Y
59082099999	24.667	113.6	Temperate	Y	Y	Y
59087099999	23.883	113.517	Temperate	-	-	Y
59096099999	24.367	114.483	Temperate	-	-	Y
59102099999	24.95	115.65	Temperate	-	-	Y
59117099999	24.283	116.067	Temperate	-	-	-
59134099999	24.544	118.128	Temperate	Y	Y	Y
59158099999	24.15	120.683	Temperate	-	-	-
59209099999	23.3	105.95	Temperate	-	-	Y
59254099999	23.4	110.083	Temperate	Y	Y	Y
59265099999	23.483	111.3	Temperate	Y	Y	Y
59278099999	23.05	112.467	Temperate	-	-	Y
59287099999	23.392	113.299	Temperate	Y	Y	Y
59293099999	23.8	114.733	Temperate	Y	Y	Y
59358099999	23.0	120.217	Tropical	-	-	-
59417099999	22.367	106.75	Temperate	Y	Y	Y

Source: data/interim/stations_analysis.csv. Coordinates are degrees north/east. Full station names, elevation, climate codes and flags are retained in the public CSV.

Table S15. Station inventory (9/9)

Analysis station IDs, coordinates, climate groups and daily-variable eligibility. Rows 321-334 of 334.

All 334 stations are used for 2-m temperature and wind speed. Tmax: 180; Tmin: 180; P24 (24-h precipitation): 249. Y = eligible; - = not eligible.

Station ID	Latitude	Longitude	Climate group	Tmax	Tmin	P24
59431099999	22.608	108.172	Temperate	Y	Y	Y
59456099999	22.35	110.933	Temperate	-	-	Y
59493099999	22.639	113.811	Temperate	-	-	-
59501099999	22.783	115.367	Temperate	Y	Y	Y
59567099999	22.033	121.55	Tropical	-	-	-
59632099999	21.95	108.617	Temperate	Y	Y	Y
59644099999	21.483	109.1	Temperate	-	-	Y
59658099999	21.15	110.3	Temperate	-	-	Y
59663099999	21.867	111.967	Temperate	Y	Y	Y
59673099999	21.733	112.767	Temperate	-	-	Y
59838099999	19.1	108.617	Tropical	Y	Y	Y
59845099999	19.517	109.583	Tropical	-	-	Y
59855099999	19.233	110.467	Tropical	Y	Y	Y
59948099999	18.217	109.583	Tropical	Y	Y	Y

Source: data/interim/stations_analysis.csv. Coordinates are degrees north/east. Full station names, elevation, climate codes and flags are retained in the public CSV.